

Problem Set 2.1.A

- (a) $x_2 - x_1 \geq 1$ or $-x_1 + x_2 \geq 1$
 (b) $x_1 + 2x_2 \geq 3$ and $x_1 + 2x_2 \leq 6$
 (c) $x_2 \geq x_1$ or $x_1 - x_2 \leq 0$
 (d) $x_1 + x_2 \geq 3$
 (e) $\frac{x_2}{x_1 + x_2} \leq .5$ or $.5x_1 - .5x_2 \geq 0$

(a) $(x_1, x_2) = (1, 4)$
 $(x_1, x_2) \geq 0$
 $6x_1 + 4x_2 = 22 < 24$
 $1x_1 + 2x_2 = 9 \neq 6$ infeasible

(b) $(x_1, x_2) = (2, 2)$
 $(x_1, x_2) \geq 0$
 $6x_2 + 4x_2 = 20 < 24$
 $1x_2 + 2x_2 = 6 = 6$
 $-1x_2 + 1x_2 = 0 < 1$
 $1x_2 = 2 = 2$ } feasible

$$Z = 5x_2 + 4x_2 = \$18$$

(c) $(x_1, x_2) = (3, 1.5)$
 $x_1, x_2 \geq 0$
 $6x_3 + 4x_{1.5} = 24 = 24$
 $1x_3 + 2x_{1.5} = 6 = 6$
 $-1x_3 + 1x_{1.5} = -1.5 < 1$
 $1x_{1.5} = 1.5 < 2$ } feasible

$$Z = 5x_3 + 4x_{1.5} = \$21$$

(d) $(x_1, x_2) = (2, 1)$
 $x_1, x_2 \geq 0$
 $6x_2 + 4x_1 = 16 < 24$
 $1x_2 + 2x_1 = 4 < 6$
 $-1x_2 + 1x_1 = -1 < 1$
 $1x_1 = 1 < 2$ } feasible

$$Z = 5x_2 + 4x_1 = \$14$$

(e) $(x_1, x_2) = (2, -1)$
 $x_1 \geq 0, x_2 < 0$, infeasible

Conclusion: (c) gives the best feasible solution

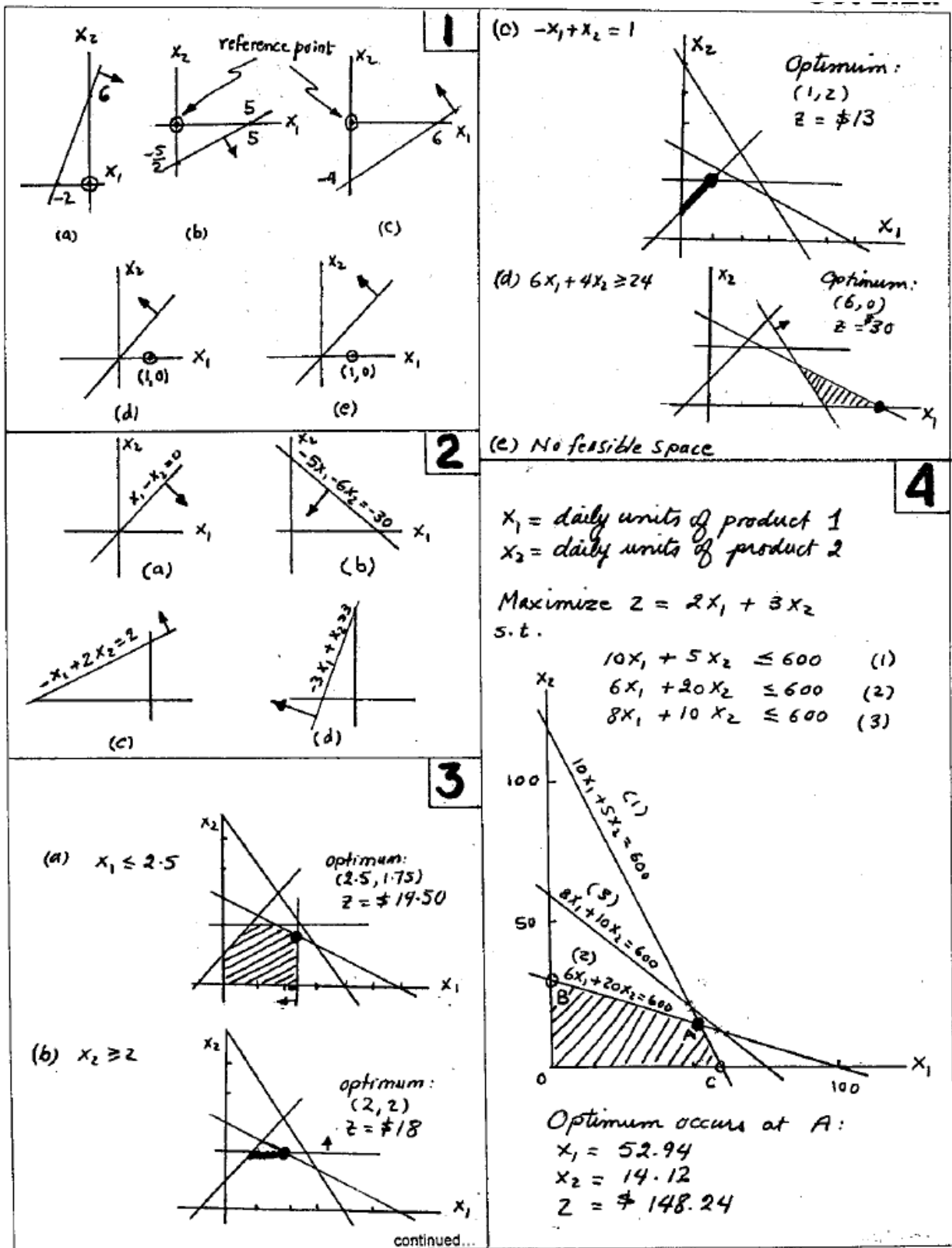
(a) $(x_1, x_2) = (2, 2)$
 Let S_1 and S_2 be the unused daily amounts of M1 and M2.
 For M1: $S_1 = 24 - (6x_1 + 4x_2) = 4$ tons/day
 For M2: $S_2 = 6 - (x_1 + 2x_2)$
 $= 6 - (2 + 2 \times 2) = 0$ tons/day

Quantity discount results in the following nonlinear objective function:

$$Z = \begin{cases} 5x_1 + 4x_2, & 0 \leq x_1 \leq 2 \\ 4.5x_1 + 4x_2, & x_1 > 2 \end{cases}$$

The situation cannot be treated as a linear program. Nonlinearity can be accounted for in this case using mixed integer programming (chapter 9).

Problem Set 2.2A



x_1 = number of units of A
 x_2 = number of units of B

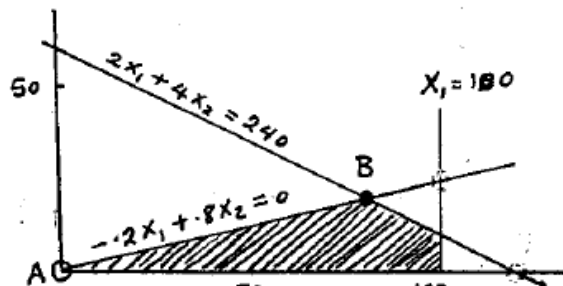
Maximize $Z = 20x_1 + 50x_2$

$$\frac{x_1}{x_1 + x_2} \geq .8 \quad \text{or} \quad -.2x_1 + .8x_2 \leq 0$$

$$x_1 \leq 100$$

$$2x_1 + 4x_2 \leq 240$$

$$x_1, x_2 \geq 0$$



Optimal occurs at B:

$$x_1 = 80 \text{ units}$$

$$x_2 = 20 \text{ units}$$

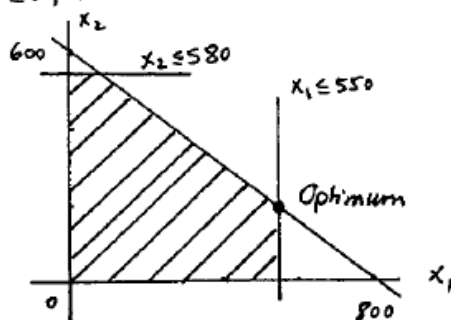
$$Z = \$2,600$$

x_1 = number of sheets/day
 x_2 = number of bars/day

Maximize $Z = 40x_1 + 35x_2$

$$\text{s.t.} \quad \frac{x_1}{800} + \frac{x_2}{600} \leq 1$$

$$0 \leq x_1 \leq 550, \quad 0 \leq x_2 \leq 580$$



Optimum solution:

$$x_1 = 550 \text{ sheets}$$

$$x_2 = 187.13 \text{ bars}$$

$$Z = \$28,549.40$$

5

x_1 = \$ invested in A

x_2 = \$ invested in B

Maximize $Z = .05x_1 + .08x_2$

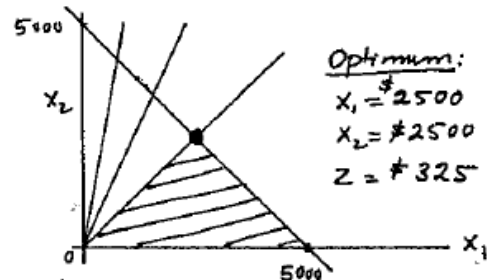
$$\text{s.t.} \quad x_1 \geq .25(x_1 + x_2)$$

$$x_2 \leq .5(x_1 + x_2)$$

$$x_1 \geq .5x_2$$

$$x_1 + x_2 \leq 5000$$

$$x_1, x_2 \geq 0$$



Optimum:

$$x_1 = \$2500$$

$$x_2 = \$2500$$

$$Z = \$325$$

x_1 = number of practical courses

x_2 = number of humanistic courses

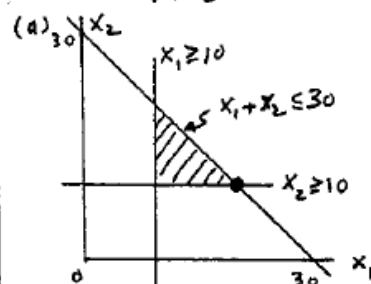
Maximize $Z = 1500x_1 + 1000x_2$

$$\text{s.t.} \quad x_1 + x_2 \leq 30$$

$$x_1 \geq 10$$

$$x_2 \geq 10$$

$$x_1, x_2 \geq 0$$



Optimum:

$$x_1 = 20$$

$$x_2 = 10$$

$$Z = \$40,000$$

(b) Change $x_1 + x_2 \leq 30$ to $x_1 + x_2 \leq 31$

$$\text{Optimum } Z = \$41,500$$

$$\Delta Z = \$41,500 - \$40,000 = \$1,500$$

Conclusion: Any additional course will be of the practical type.

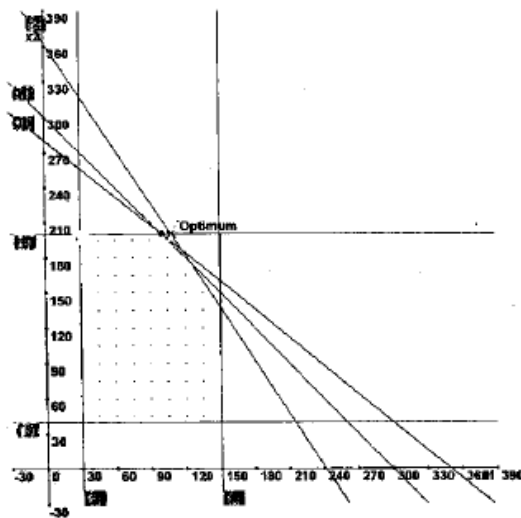
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8

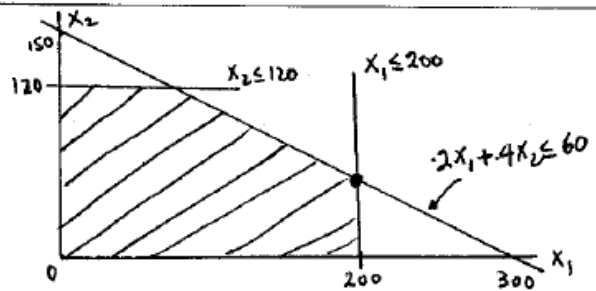
x_1 = units of solution A
 x_2 = units of solution B
 Maximize $Z = 8x_1 + 10x_2$
 Subject to
 $5x_1 + 5x_2 \leq 150$
 $6x_1 + 4x_2 \leq 145$
 $x_1 \geq 30$
 $x_1 \leq 150$
 $x_2 \geq 40$
 $x_2 \leq 200$
 $x_1, x_2 \geq 0$

Summary of Optimal Solution:
 Objective Value = 2800.00
 $x_1 = 100.00$
 $x_2 = 200.00$



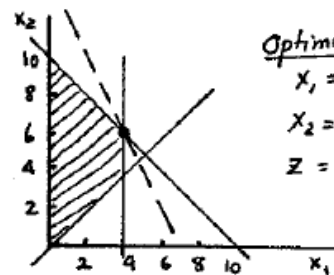
x_1 = nbr. of grano boxes
 x_2 = nbr. of wheatie boxes
 Maximize $Z = x_1 + 1.35x_2$
 s.t. $2x_1 + 4x_2 \leq 60$
 $x_1 \leq 200$
 $x_2 \leq 120$
 $x_1, x_2 \geq 0$

9



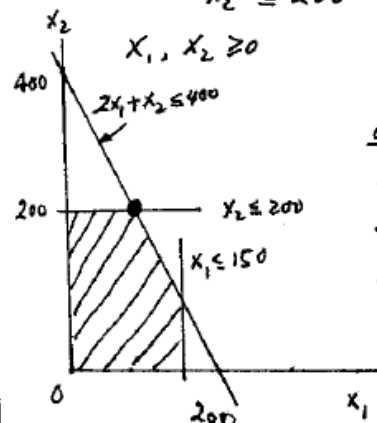
Optimum: $x_1 = 200$, $x_2 = 50$, $Z = \$267.50$
 Area allocation: 67% grano, 33% wheatie

x_1 = play hours per day
 x_2 = work hours per day
 Maximize $Z = 2x_1 + x_2$
 s.t.
 $x_1 + x_2 \leq 10$
 $x_1 - x_2 \leq 0$
 $x_1 \leq 4$
 $x_1, x_2 \geq 0$



Optimum solution:
 $x_1 = 4$ hours
 $x_2 = 6$ hours
 $Z = 14$ "pleasurits"

x_1 = Daily nbr. of type 1 hat
 x_2 = Daily nbr. of type 2 hat
 Maximize $Z = 8x_1 + 5x_2$
 s.t. $2x_1 + x_2 \leq 400$
 $x_1 \leq 150$
 $x_2 \leq 200$
 $x_1, x_2 \geq 0$



Optimum:
 $x_1 = 100$ type 1
 $x_2 = 200$ type 2
 $Z = \$1800$

continued...

continued...

x_1 = radio minutes

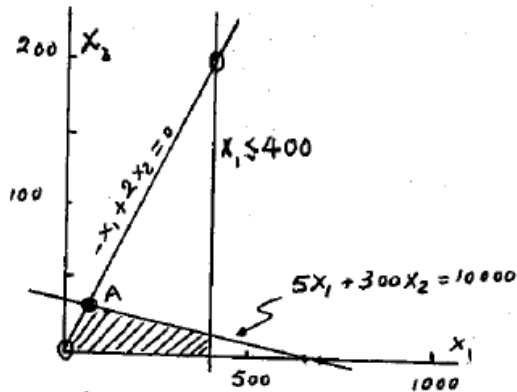
x_2 = TV minutes

Maximize $Z = x_1 + 25x_2$

s.t. $15x_1 + 300x_2 \leq 10,000$

$$\frac{x_1}{x_2} \geq Z \text{ or } -x_1 + 2x_2 \leq 0$$

$$x_1 \leq 400, x_1, x_2 \geq 0$$



Optimum occurs at A:

$$x_1 = 60.61 \text{ minutes}$$

$$x_2 = 30.3 \text{ minutes}$$

$$Z = 818.18$$

13

(a) Optimum occurs at A:

$$x_1 = 5.128 \text{ tons per hour}$$

$$x_2 = 10.256 \text{ tons per hour}$$

$$Z = 153,846 \text{ lb of steam}$$

$$\text{Optimal ratio} = \frac{5.128}{10.256} = .5$$

(b) $2.1x_1 + .9x_2 \leq (20+1) = 21$

$$\text{Optimum } Z = 16,538 \text{ lb of steam}$$

$$\Delta Z = 16,538 - 153,846 = 7692 \text{ lb}$$

x_1 = tons of C_1 consumed per hour

x_2 = tons of C_2 consumed per hour

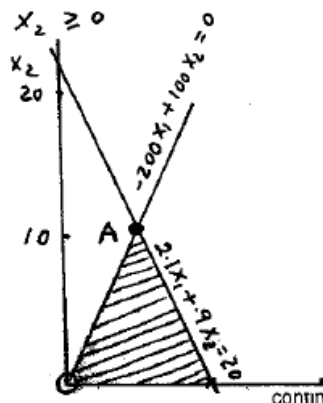
Maximize $Z = 12000x_1 + 9000x_2$

s.t. $1800x_1 + 2100x_2 \leq 2000(x_1 + x_2)$

$$\text{or } -200x_1 + 100x_2 \leq 0$$

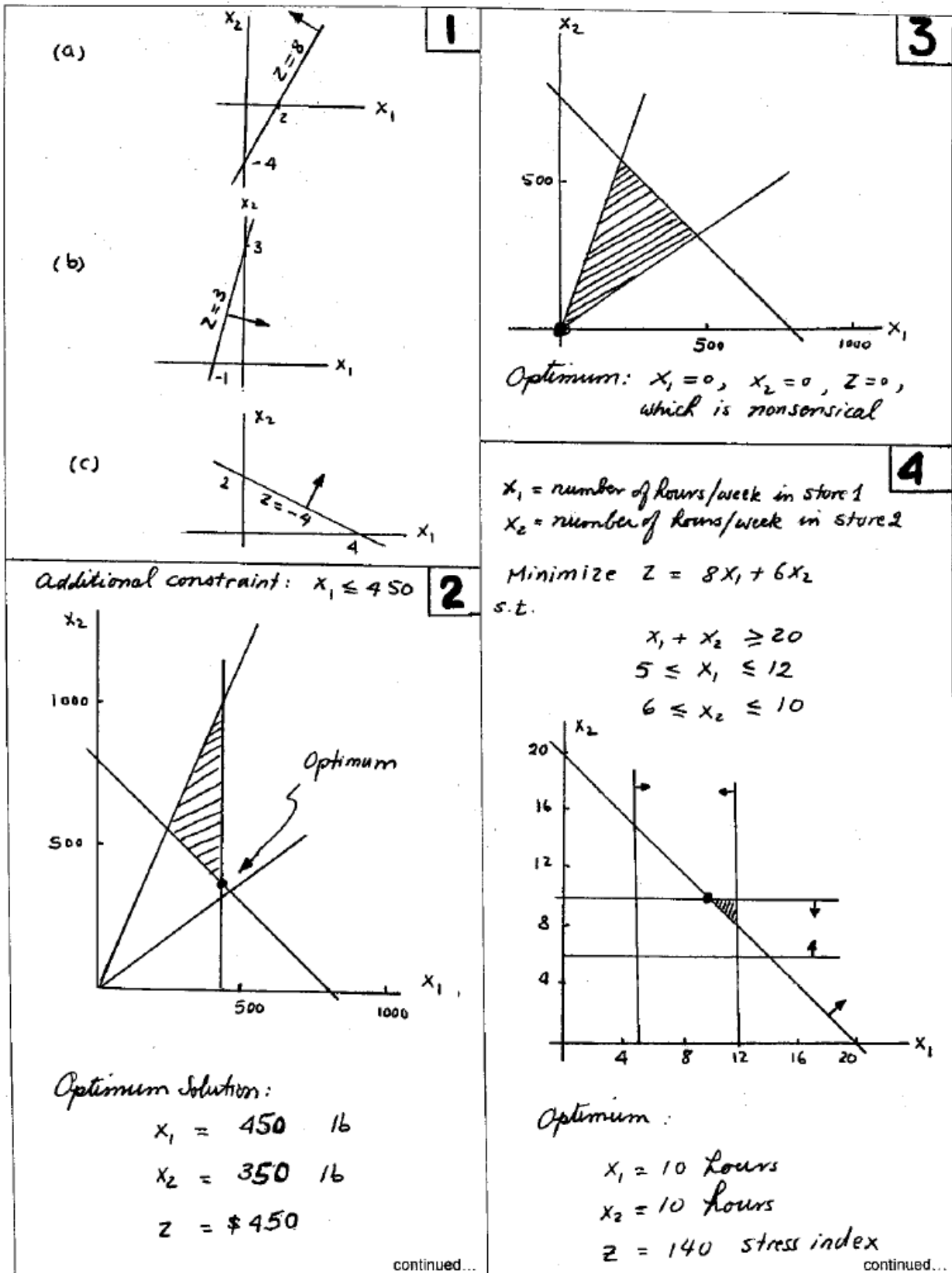
$$2.1x_1 + .9x_2 \leq 20$$

$$x_1, x_2 \geq 0$$



14

Problem Set 2.2B



5

Let

 $x_1 = 10^3$ bbl/day from Iran $x_2 = 10^3$ bbl/day from DubaiRefinery capacity = $x_1 + x_2 \leq 10^3$ bbl/dayMinimize $Z = x_1 + x_2$

Subject to

$$x_1 \geq .4(x_1 + x_2)$$

$$\text{or } -.6x_1 + .4x_2 \leq 0$$

$$.2x_1 + .1x_2 \geq 14$$

$$.25x_1 + .6x_2 \geq 30$$

$$.1x_1 + .15x_2 \geq 10$$

$$.15x_1 + .1x_2 \geq 8$$

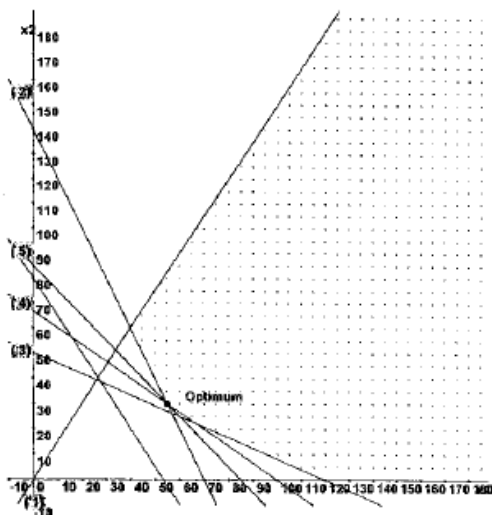
$$x_1, x_2 \geq 0$$

Optimum solution from TORA:

LINEAR PROGRAMMING - GRAPHICAL SOLUTION

Title: diet problem

Summary of Optimal Solution:
 Objective Value = 85.00
 $x_1 = 65.00$
 $x_2 = 30.00$



6

Let

 $x_1 = 10^3$ invested in blue chip stock $x_2 = 10^3$ invested in high-tech stocksMinimize $Z = x_1 + x_2$

Subject to

$$.1x_1 + .25x_2 \geq 10$$

$$.6x_1 - .4x_2 \geq 0$$

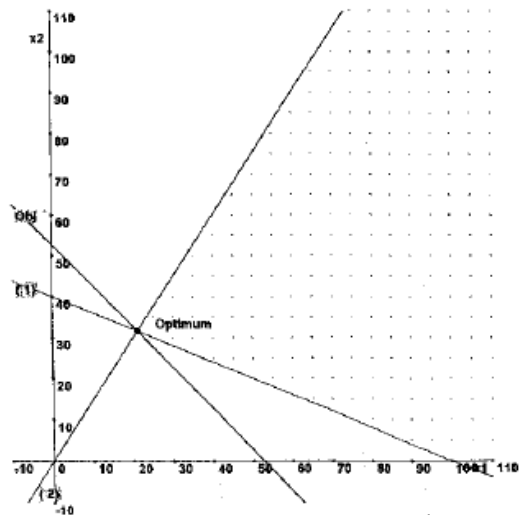
$$x_1, x_2 \geq 0$$

TORA optimum solution:

LINEAR PROGRAMMING - GRAPHICAL SOLUTION

Title: diet problem

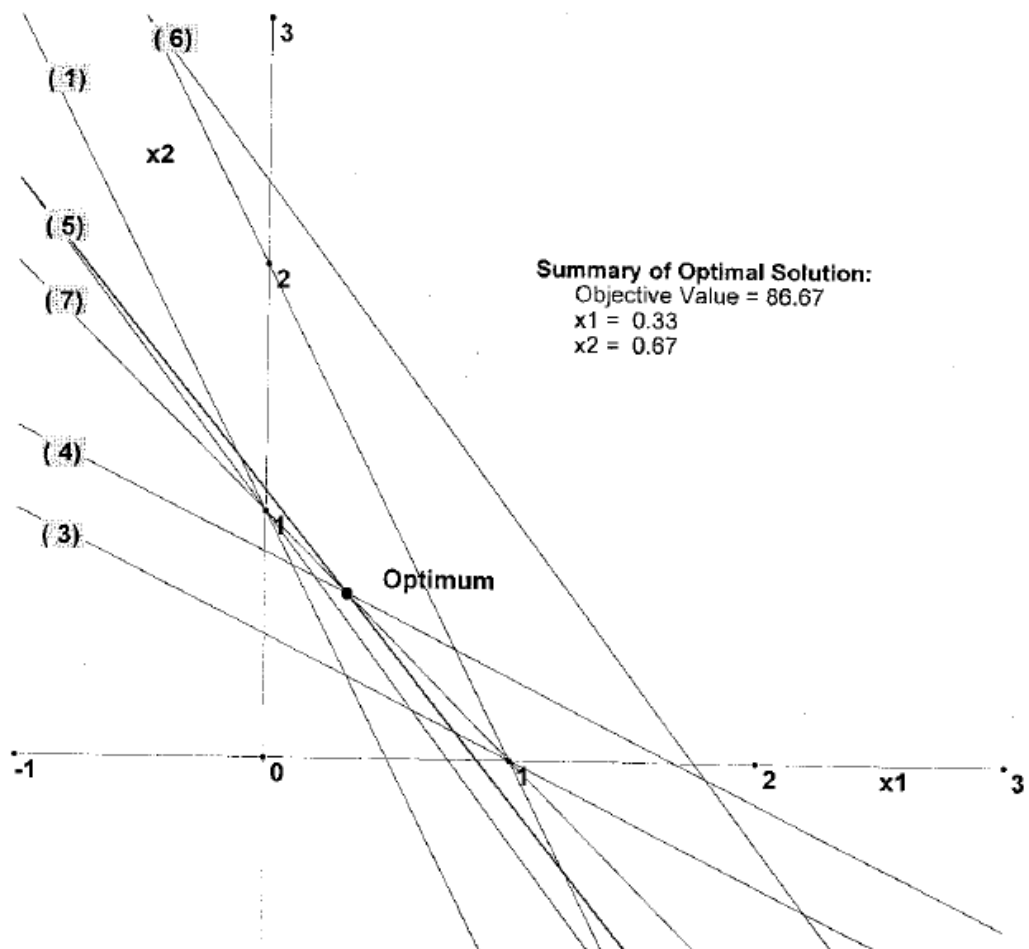
Summary of Optimal Solution:
 Objective Value = 52.63
 $x_1 = 25.05$
 $x_2 = 31.58$



7

x_1 = Ratio of scrap A in alloy
 x_2 = Ratio of scrap B in alloy

	x_1	x_2		
Minimize	100.00	80.00		
Subject to				
(1)	0.06	0.03	$\geq$	0.03
(2)	0.06	0.03	$\leq$	0.06
(3)	0.03	0.06	$\geq$	0.03
(4)	0.03	0.06	$\leq$	0.05
(5)	0.04	0.03	$\geq$	0.03
(6)	0.04	0.03	$\leq$	0.07
(7)	1.00	1.00	$=$	1.00



Problem Set 3.1A

$(x_1, x_2) = (3, 1)$ M1: $S_1 = 24 - (6 \times 3 + 4 \times 1) = 2$ tons/day M2: $S_2 = 6 - (1 \times 3 + 2 \times 1) = 1$ ton/day	1 $y = \max \{ x_1 - x_2 + 3x_3 , -x_1 + 3x_2 - x_3 \}$ 5
$S_1 = x_1 + x_2 - 800$ $= 500 + 600 - 800 = 300$ lb	2 Hence $x_1 - x_2 + 3x_3 \leq y$ $-x_1 + 3x_2 - x_3 \leq y$
$10x_1 - 3x_2 \geq -5 \equiv -10x_1 + 3x_2 \leq 5$ Thus, $-10x_1 + 3x_2 + S_1 = 5$ ① Also, $10x_1 - 3x_2 \geq -5 \equiv 10x_1 - 3x_2 - S_2 = -5$ Thus, $-10x_1 + 3x_2 + S_2 = 5$ ② ① and ② are the same	3 LP model: minimize $Z = y$ Subject to
x_{ij} = number of units of product i manufactured on machine j 4 LP model Maximize $Z = 10(x_{11} + x_{12}) + 15(x_{21} + x_{22})$ Subject to $(x_{11} + x_{21}) - (x_{12} + x_{22}) \leq 5$ $x_{11} + x_{21} \leq 200$ $x_{12} + x_{22} \leq 250$ $x_{ij} \geq 0$ for all $i \neq j$	$x_1 - x_2 + 3x_3 \leq y$ $x_1 - x_2 + 3x_3 \geq -y$ $-x_1 + 3x_2 - x_3 \leq y$ $-x_1 + 3x_2 - x_3 \geq -y$ $x_1, x_2, x_3, y \geq 0$
Equation form: $(x_{11} + x_{21}) - (x_{12} + x_{22}) \leq 5$ to $x_{11} + x_{21} - x_{12} - x_{22} \leq 5$ $x_{11} + x_{21} - x_{12} - x_{22} \geq -5$ Maximize $Z = 10x_{11} + 10x_{12} + 15x_{21} + 15x_{22}$ Subject to $x_{11} + x_{21} - x_{12} - x_{22} + S_1 = 5$ $-x_{11} - x_{21} + x_{12} + x_{22} + S_2 = -5$ $x_{11} + x_{21} + S_3 = 200$ $x_{12} + x_{22} + S_4 = 250$ $x_{ij} \geq 0$ for all i and j $S_i \geq 0$ for all i continued...	Equation form: Minimize $Z = y$ Subject to $-y + x_1 - x_2 + 3x_3 + S_1 = 0$ $-y - x_1 + x_2 - 3x_3 + S_2 = 0$ $-y - x_1 + 3x_2 - x_3 + S_3 = 0$ $-y + x_1 - 3x_2 + x_3 + S_4 = 0$ $x_1, x_2, x_3, y, S_1, S_2, S_3, S_4 \geq 0$

Problem Set 3.1B

1

$x_1 = \text{Nbr. } \frac{1}{4}\text{-lb / day}$
 $x_2 = \text{Nbr. cheeseburgers/day}$
 Maximize $Z = .2x_1 + .15x_2 - .25x_3^+$
 s.t.
 $.25x_1 + .2x_2 + x_3^- - x_3^+ = 200$
 $x_1 + x_2 \leq 900$
Solution: $Z = \$173.35$
 $x_1 = 900, x_2 = 0, x_3^+ = 25 \text{ lb}$

2

$x_j = \# \text{ units of product } j \text{ per day, } j=1,2$
 $x_3^- = \text{unused minutes of machine time/day}$
 $x_3^+ = \text{machine overtime per day in minutes}$
 Maximize $Z = 6x_1 + 7.5x_2 - .5x_3^-$
 Subject to
 $10x_1 + 12x_2 + x_3^- - x_3^+ = 2500$
 $150 \leq x_1 \leq 200$
 $x_2 \leq 45$
 $x_1, x_2 \geq 0$
 $x_3^+, x_3^- \geq 0$
TORA optimum solution:
 $x_1 = 200 \text{ units/day}$
 $x_2 = 45 \text{ units/day}$
 $x_3^+ = \text{overtime minutes}$
 $= 40 \text{ minutes/day}$
 $Z = \$1517.50$

3

$x_j = \# \text{ units of products } 1, 2, \text{ and } 3$
 Maximize $Z = 2x_1 + 5x_2 + 3x_3 - 15x_4^+ - 10x_5^+$
 Subject to
 $2x_1 + x_2 + 2x_3 + x_4^- - x_4^+ = 80$
 $x_1 + x_2 + 2x_3 + x_5^- - x_5^+ = 65$
 all variables ≥ 0
Solution: $Z = \$325$
 $x_2 = 65 \text{ units, } x_4^- = 15$
 All other variables = 0

continued...

4

Formulation 1:
 Maximize $Z = -2x_1 + 3x_2^+ - 3x_2^- - 2x_3^+ + 2x_3^-$
 Subject to
 $4x_1 - x_2^+ + x_2^- - 5x_3^+ + 5x_3^- = 10$
 $2x_1 + 3x_2^+ - 3x_2^- + 2x_3^+ - 2x_3^- = 12$
 all variables ≥ 0
Optimum solution:
 $x_1 = 0$
 $x_2^+ = 6.15$
 $x_2^- = 0$
 $x_3^+ = 0$
 $x_3^- = 3.23$
 $Z = 24.92$

Formulation 2:
 Maximize $Z = -2x_1 + 3x_2^+ - 2x_3^+ - w$
 Subject to
 $4x_1 - x_2^+ - 5x_3^+ + 6w = 10$
 $2x_1 + 3x_2^+ + 2x_3^+ - 5w = 12$
 all variables ≥ 0
Optimum solution:
 $x_1 = 0$
 $x_2^+ = 9.38$
 $w = 3.23$
 $x_3^+ = 0$
 $w = 3.23$
 $Z = 24.92$

continued...

Problem Set 3.2A

(a)

Equation form:

$$\begin{aligned} \text{Maximize } Z &= 2x_1 + 3x_2 \\ \text{Subject to } & x_1 + 3x_2 + x_3 = 6 \\ & 3x_1 + 2x_2 + x_4 = 6 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

(b) Basic (x_1, x_2) (Point B):

$$\begin{aligned} x_1 + 3x_2 &= 6 \\ 3x_1 + 2x_2 &= 6 \\ \text{Solution: } (x_1, x_2) &= \left(\frac{6}{7}, \frac{12}{7}\right), Z = 6\frac{6}{7} \end{aligned}$$

Basic (x_1, x_3) (Point E):

$$\begin{aligned} x_1 + x_3 &= 6 \\ 3x_1 &= 6 \\ \text{Solution: } (x_1, x_3) &= (2, 4), Z = 4 \end{aligned}$$

Basic (x_1, x_4) (Point C):

$$\begin{aligned} x_1 &= 6 \\ 3x_1 + x_4 &= 6 \\ \text{Solution: } (x_1, x_4) &= (6, -12) \\ &\text{Unique but infeasible} \end{aligned}$$

Basic (x_2, x_3) (Point A):

$$\begin{aligned} 3x_2 + x_3 &= 6 \\ 2x_2 &= 6 \\ \text{Solution: } (x_2, x_3) &= (3, -3) \\ &\text{Unique but infeasible} \end{aligned}$$

Basic (x_2, x_4) (Point D):

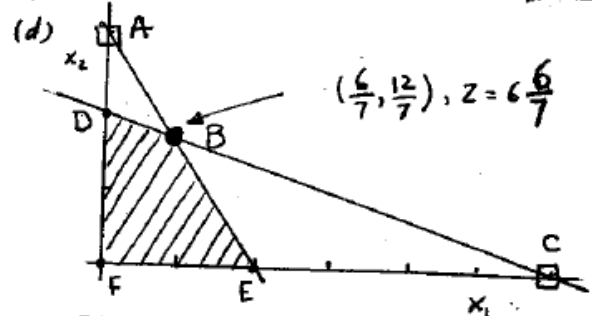
$$\begin{aligned} 3x_2 &= 6 \\ 2x_2 + x_4 &= 6 \\ \text{Solution: } (x_2, x_4) &= (2, 2), Z = 6 \end{aligned}$$

Basic (x_3, x_4) (Point F):

$$\begin{aligned} x_3 &= 6 \\ x_4 &= 6 \\ \text{Solution: } (x_3, x_4) &= (6, 6), Z = 0 \end{aligned}$$

(c) Optimum solution occurs at B:
 $(x_1, x_2) = \left(\frac{6}{7}, \frac{12}{7}\right)$ with $Z = 6\frac{6}{7}$

continued...



(e) From the graph in (d), we have

$$A: x_2 = 3, x_3 = -3$$

$$C: x_1 = 6, x_4 = -12$$

(a) Maximize $Z = 2x_1 - 4x_2 + 5x_3 - 6x_4$
Subject to

$$\begin{aligned} x_1 + 4x_2 - 2x_3 + 8x_4 + x_5 &= 2 \\ -x_1 + 2x_2 + 3x_3 + 4x_4 + x_6 &= 1 \\ x_1, x_2, x_3, x_4, x_5, x_6 &\geq 0 \end{aligned}$$

Combination	Solution	Status	Z
x_1, x_2	0, 1/2	Feasible	-2
x_1, x_3	8, 3	Feasible	31
x_1, x_4	0, 1/4	Feasible	-3/2
x_1, x_5	-1, 3	Infeasible	—
x_1, x_6	2, 3	Feasible	4
x_2, x_3	1/2, 0	Feasible	-2
x_2, x_4	1/2, 0	Feasible	-2
x_2, x_5	1/2, 0	Feasible	-2
x_2, x_6	1/2, 0	Feasible	-2
x_3, x_4	0, 1/4	Feasible	-3/2
x_3, x_5	1/3, 8/3	Feasible	5/3
x_3, x_6	-1, 4	Infeasible	—
x_4, x_5	1/4, 0	Feasible	-3/2
x_4, x_6	1/4, 0	Feasible	-3/2
x_5, x_6	2, 1	Feasible	0

Optimum Solution:

$$x_1 = 8, x_2 = 0, x_3 = 3, x_4 = 0$$

$$Z = 31$$

continued...

(b) Minimize $Z = x_1 + 2x_2 - 3x_3 - 2x_4$
 Subject to

$$x_1 + 2x_2 - 3x_3 + x_4 = 4$$

$$x_1 + 2x_2 + x_3 + 2x_4 = 4$$

$$x_1, x_2, x_3, x_4 \geq 0$$

Combination Solution Status Z

x_1, x_2	infinity of solutions	—
x_1, x_3	4, 0	Feasible 4
x_1, x_4	4, 0	Feasible 4
x_2, x_3	2, 0	Feasible 4
x_2, x_4	2, 0	Feasible 4
x_3, x_4	$-\frac{4}{7}, \frac{16}{7}$	Infeasible —

Alternative optima:

x_1	x_2	x_3	x_4	Z
4	0	0	0	4
0	2	0	0	4

Maximize $Z = 2x_1 + 3x_2^- - 3x_2^+ + 5x_3$

Subject to

$$-6x_1 + 7x_2^- - 7x_2^+ - 9x_3 - x_4 = 4$$

$$x_1 + x_2^- - x_2^+ + 4x_3 = 10$$

$$x_1, x_2^-, x_2^+, x_3, x_4 \geq 0$$

(x_2^-, x_2^+) :

$$7x_2^- - 7x_2^+ = 4$$

$$x_2^- - x_2^+ = 10$$

Since $(7x_2^- - 7x_2^+)$ and $(x_2^- - x_2^+)$ are dependent, it is impossible for x_2^- and x_2^+ to be basic simultaneously. This means that at least x_2^- and x_2^+ must be nonbasic at zero level; thus making it impossible for x_2^- and x_2^+ to assume positive values simultaneously in any basic solution.

Maximize $Z = x_1 + x_2$
 Subject to

$$x_1 + 2x_2 + x_3 = 6$$

$$2x_1 + x_2 - x_4 = 16$$

$$x_1, x_2, x_3, x_4 \geq 0$$

Combination Solution Status

x_1, x_2	$26/3, -4/3$	Infeasible
x_1, x_3	8, -2	Infeasible
x_1, x_4	6, -4	Infeasible
x_2, x_3	16, -26	Infeasible
x_2, x_4	3, -13	Infeasible
x_3, x_4	6, -16	Infeasible

Maximize $Z = x_1 + 3x_2$
 Subject to

$$x_1 + x_2 + x_3 = 2$$

$$-x_1 + x_2 + x_4 = 4$$

$$x_1 \text{ unrestricted}$$

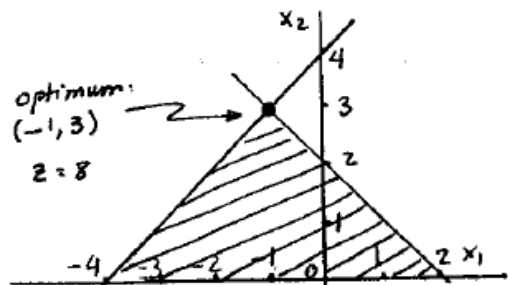
$$x_2, x_3 \geq 0$$

Combination Solution Status Z

x_1, x_2	-1, 3	Feasible	8
x_1, x_3	-4, 6	Feasible	-4
x_1, x_4	2, 6	Feasible	2
x_2, x_3	4, -2	Infeasible	—
x_2, x_4	2, 2	Feasible	6
x_3, x_4	2, 4	Feasible	0

Optimum: $x_1 = -1, x_2 = 3, Z = 8$

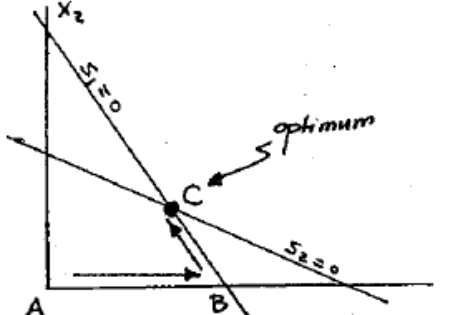
(c)



continued...

Problem Set 3.3A

1

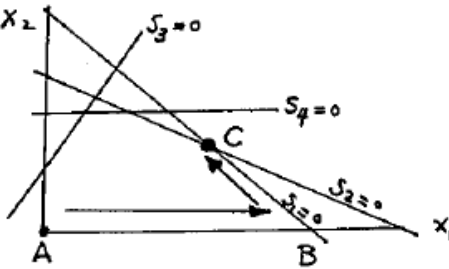


Extreme point	Basic	Nonbasic
A	S_1, S_2	x_1, x_2
B	x_1, S_2	x_2, S_1
C	x_1, x_2	S_1, S_2

4

Extreme Point	Basic	Nonbasic
A	S_1, S_2, S_3, S_4	x_1, x_2, x_3
B	S_1, x_1, S_3, S_4	S_2, x_2, S_3
C	x_2, S_2, S_3, S_4	S_1, x_1, x_3
D	S_1, S_2, x_3, S_4	x_1, x_2, S_3
E	x_1, x_2, S_3, S_4	S_1, S_2, x_3
F	x_2, S_2, x_3, S_4	x_1, S_1, S_3
G	S_1, x_1, x_3, S_4	S_2, x_2, S_3
H	S_1, x_1, x_2, x_3	S_2, S_3, S_4
I	x_1, x_2, x_3, S_3	S_1, S_2, S_4
J	x_1, S_2, x_2, x_3	S_1, S_3, S_4

2



Extreme point	Basic	Nonbasic
A	S_1, S_2, S_3, S_4	x_1, x_2
B	x_1, S_2, S_3, S_4	S_1, x_2
C	x_1, x_2, S_3, S_4	S_1, S_2

5

(a) x_3 enters at value 1
 $Z = 0 + 3 \times 1 = 3$

(b) x_1 enters at value 1
 $Z = 0 + 5 \times 1 = 5$

(c) x_2 enters at value 1
 $Z = 0 + 7 \times 1 = 7$

(d) Tie broken arbitrarily between x_1, x_2 , and x_3 . Entering value = 1
 $Z = 0 + 1 \times 1 = 1$

3

(a) (A, B) adjacent, hence can be on a simplex path. Remaining pairs cannot be on a simplex path because they are not adjacent.

(b) (i) Yes, because connects adjacent extreme points
 (ii) No, because A and E are not adjacent.
 (iii) No, because the path returns to a previous extreme point.

Problem Set 3.3B

Basic	Z	X ₁	X ₂	S ₁	S ₂	S ₃	S ₄	Sol
Z	1	-5	-4	0	0	0	0	0
S ₁	0	6	4	1	0	0	0	24
S ₂	0	1	2	0	1	0	0	6
S ₃	0	-1	1	0	0	1	0	1
S ₄	0	0	1	0	0	0	1	2
Z	1	0	6	0	5	0	0	30
S ₁	0	0	-8	1	-6	0	0	12
X ₁	0	1	2	0	1	0	0	6
S ₃	0	0	3	0	1	1	0	7
S ₄	0	0	1	0	0	0	1	2

(a)

Basic	x ₁	x ₂	x ₃	x ₄	sx ₅	sx ₆	sx ₇	Solution
Z	-2.00	1.00	3.00	-5.00	0.00	0.00	0.00	0.00
1)ax5	1.00	2.00	2.00	4.00	1.00	0.00	0.00	40.00
2)ax6	2.00	-1.00	1.00	2.00	0.00	1.00	0.00	8.00
3)ax7	4.00	-2.00	1.00	-1.00	0.00	0.00	1.00	10.00
Z	3.00	-3.50	5.50	0.00	0.00	2.50	0.00	20.00
1)ax5	-3.00	4.00	0.00	0.00	1.00	-2.00	0.00	24.00
2)ax6	1.00	-0.50	0.50	1.00	0.00	0.50	0.00	4.00
3)ax7	5.00	-2.50	1.50	0.00	0.00	0.50	1.00	14.00
Z	0.30	0.00	5.50	0.00	0.88	0.75	0.00	41.00
1)ax2	-0.75	1.00	0.00	0.00	0.25	-0.50	0.00	6.00
2)ax4	0.62	0.00	0.50	1.00	0.12	0.25	0.00	7.00
3)ax7	3.12	0.00	1.50	0.00	0.62	-0.75	1.00	29.00

(b)

Basic	x ₁	x ₂	x ₃	x ₄	sx ₅	sx ₆	sx ₇	Solution
Z	-8.00	-6.00	-3.00	2.00	0.00	0.00	0.00	0.00
1)ax5	1.00	2.00	2.00	4.00	1.00	0.00	0.00	40.00
2)ax6	2.00	-1.00	1.00	2.00	0.00	1.00	0.00	8.00
3)ax7	4.00	-2.00	1.00	-1.00	0.00	0.00	1.00	10.00
Z	0.00	-10.00	-1.00	0.00	0.00	0.00	2.00	20.00
1)ax5	0.00	2.50	1.75	4.25	1.00	0.00	-0.25	37.00
2)ax6	0.00	0.00	0.50	2.50	0.00	1.00	-0.50	3.00
3)ax1	1.00	-0.50	0.25	-0.25	0.00	0.00	0.25	2.50
Z	0.00	0.00	6.00	17.00	4.00	0.00	1.00	170.00
1)ax2	0.00	1.00	0.70	1.70	0.40	0.00	-0.10	15.00
2)ax4	0.00	0.00	0.50	2.50	0.00	1.00	-0.50	3.00
3)ax1	1.00	0.00	0.60	0.60	0.20	0.00	0.20	10.00

(c)

Basic	x ₁	x ₂	x ₃	x ₄	sx ₅	sx ₆	sx ₇	Solution
Z	-3.00	1.00	-3.00	-4.00	0.00	0.00	0.00	0.00
1)ax5	1.00	2.00	2.00	4.00	1.00	0.00	0.00	40.00
2)ax6	2.00	-1.00	1.00	2.00	0.00	1.00	0.00	8.00
3)ax7	4.00	-2.00	1.00	-1.00	0.00	0.00	1.00	10.00
Z	1.00	-1.00	-1.00	0.00	0.00	2.00	0.00	16.00
1)ax5	-3.00	4.00	0.00	0.00	1.00	-2.00	0.00	24.00
2)ax4	1.00	-0.50	0.50	1.00	0.00	0.50	0.00	4.00
3)ax7	5.00	-2.50	1.50	0.00	0.00	0.50	1.00	14.00
Z	0.25	0.00	-1.00	0.00	0.25	1.50	0.00	22.00
1)ax2	-0.75	1.00	0.00	0.00	0.25	-0.50	0.00	6.00
2)ax4	0.62	0.00	0.50	1.00	0.12	0.25	0.00	7.00
3)ax7	3.12	0.00	1.50	0.00	0.62	-0.75	1.00	29.00
Z	1.50	0.00	0.00	2.00	0.50	2.00	0.00	36.00
1)ax2	-0.75	1.00	0.00	0.00	0.25	-0.50	0.00	6.00
2)ax5	1.25	0.00	1.00	2.00	0.25	0.50	0.00	16.00
3)ax7	1.25	0.00	0.00	-3.00	0.25	-1.50	1.00	8.00

continued...

Basic	x ₁	x ₂	x ₃	x ₄	sx ₅	sx ₆	sx ₇	Solution
Z	-5.00	4.00	-6.00	8.00	0.00	0.00	0.00	0.00
1)ax5	1.00	2.00	2.00	4.00	1.00	0.00	0.00	40.00
2)ax6	2.00	-1.00	1.00	2.00	0.00	1.00	0.00	8.00
3)ax7	4.00	-2.00	1.00	-1.00	0.00	0.00	1.00	10.00
Z	-13.00	8.00	-10.00	0.00	0.00	-4.00	0.00	-32.00
1)ax5	-3.00	4.00	0.00	0.00	1.00	-2.00	0.00	24.00
2)ax4	1.00	-0.50	0.50	1.00	0.00	0.50	0.00	4.00
3)ax7	5.00	-2.50	1.50	0.00	0.00	0.50	1.00	14.00
Z	-7.00	0.00	-10.00	0.00	-2.00	0.00	0.00	-60.00
1)ax2	-0.75	1.00	0.00	0.00	0.25	-0.50	0.00	6.00
2)ax4	0.62	0.00	0.50	1.00	0.12	0.25	0.00	7.00
3)ax7	3.12	0.00	1.50	0.00	0.62	-0.75	1.00	29.00

Basic	Ratios			
	X ₁	X ₂	X ₃	X ₄
X ₅	4/1	4/2	--	(4/5)
X ₆	8/5	--	--	8/6
X ₇	3/2	3/3	--	3/3
X ₈	--	--	(0/1)	--
Value	1.5	1	0	0.8
Leaving var	X ₇	X ₇	X ₈	X ₅

(a) Nonbasic x₁ will improve solution.Basic x₁-ratios

$$\begin{array}{l} X_2 \quad (4/5) \Rightarrow X_2 \text{ leaves, } X_1 = \frac{4}{5} \\ X_3 \quad 8/6 \\ X_4 \quad 3/3 \end{array}$$

$$X_1 = \frac{4}{5} = 0.8, X_3 = 8 - 6 \times 0.8 = 3.6, X_4 = 3 - 3 \times 0.8 = 0.6$$

(b) x₁ remains nonbasic at zero. Current solution, x₂ = 4, x₃ = 8, x₄ = 3, Z = 0 is optimum

Basic solutions consist of one variable each. Thus,

$$\begin{array}{l} X_1 = 90/1 = 90, \quad Z = 5 \times 90 = 450 \\ X_2 = 90/3 = 30, \quad Z = -6 \times 30 = -180 \\ X_3 = 90/5 = 18, \quad Z = 3 \times 18 = 54 \\ X_4 = 90/6 = 15, \quad Z = -5 \times 15 = -75 \\ X_5 = 90/3 = 30, \quad Z = 12 \times 30 = 360 \end{array}$$

Optimum solution:

$$X_1 = 90, X_2 = X_3 = X_4 = X_5 = 0, Z = 450$$

(a) Basic: (x₂, x₃, x₁) = (12, 6, 0), Z = 620Nonbasic: (x₄, x₅, x₆, x₇) = (0, 0, 0, 0)(b) x₂, x₅, x₆ will improve solution.

$$X_2 \text{ enters: } X_2 = \min\left(\frac{12}{3}, \frac{6}{1}, -\right) = 4. \text{ Thus, } X_8 \text{ leaves, } \Delta Z = 4 \times 5 = 20$$

continued...

X_5 enters: $X_5 = \min(-, \frac{6}{1}, \frac{0}{6}) = 0$. Thus, $\Delta Z = 1 \times 0 = 0$ (X_1 leaves)

X_6 enters: $X_6 = \min(-, -, -)$. Thus, no leaving variable and X_6 can be increased to ∞ . $\Delta Z = +\infty$

(c) X_4 can improve solution.

X_4 enters: $X_4 = \min(-, \frac{6}{3}, -) = 2$. Thus, X_3 leaves. $\Delta Z = -4 \times 2 = -8$

(d) As shown in (b), X_5 cannot change Z because it enters the solution at level zero. X_7 cannot change Z either because its objective equation coefficient = 0. $\Delta Z = 0 \times \min(\frac{12}{5}, \frac{6}{3}, -) = 0$

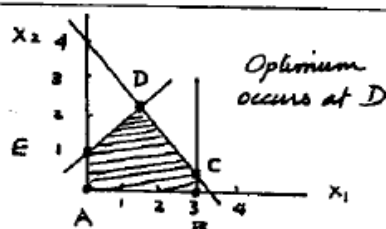
(a) Maximize $Z = 3X_1 + 6X_2$:
 X_2 is the first entering variable.
Resulting path is $A \rightarrow G \rightarrow F \rightarrow E$.

(b) Maximize $Z = 4X_1 + X_2$:
Entering variable $X_1 = (\text{min intercept with } X_1\text{-axis})$

$$X_1 = \min(2, 3, 5) = 2 \text{ at B}$$

$$\Delta Z = 4 \times 2 = 8$$

(c) Maximize $Z = X_1 + 4X_2$:
Entering variable $X_2 = (\text{min intercept with } X_2\text{-axis})$
 $X_2 = \min(1, 2, 4) = 1$
 $\Delta Z = 4 \times 1 = 4$



(a) X_1 will enter first and the iterations will follow the path $A \rightarrow B \rightarrow C \rightarrow D$.
(b) X_2 enters first and the iterations will follow the path $A \rightarrow E \rightarrow D$.
(c) The most-negative criterion requires more iterations (4 vs. 3). This criterion is only a heuristic, and although it does not guarantee the smallest number of

continued...

iterations, computational experience demonstrates that, on the average, the most-negative criterion is more efficient.

(d) Iterations are identical, with the exception of the objective row, which should appear with an opposite sign.

Optimum tableau:

Basic	X_1	X_2	S_1	S_2	S_3	S_4	
Z	0	0	$\frac{3}{4}$	$\frac{1}{2}$	0	0	21
X_1	1	0	$\frac{1}{4}$	$-\frac{1}{2}$	0	0	3
X_2	0	1	$-\frac{1}{8}$	$\frac{3}{4}$	0	0	$\frac{3}{2}$
S_3	0	0	$\frac{3}{8}$	$-\frac{5}{4}$	1	0	$\frac{5}{2}$
S_4	0	0	$\frac{1}{8}$	$-\frac{3}{4}$	0	1	$\frac{1}{2}$

If S_1 enters, its value = $\min\{\frac{3}{1/4}, -\frac{5/2}{3/8}, \frac{1/2}{1/8}\} = 4$
New $Z = 21 - 3/4 \times 4 = 18$

If S_2 enters, its value = $\min\{-\frac{3/2}{3/4}, -\frac{5/2}{-5/4}, -\frac{1/2}{-3/4}\} = 2$
New $Z = 21 - 4 \times 2 = 20$. The second best Z is associated with S_2 entering the basic solution.

Not easily extendable because the third best solution may not be an adjacent corner point of the current optimum point.

X_1 = number of pumas per day
 X_2 = number of bags per day
 X_3 = number of backpacks per day

Maximize $Z = 24X_1 + 22X_2 + 45X_3$
Subject to

$$\begin{aligned} 2X_1 + X_2 + 3X_3 &\leq 42 \\ 2X_1 + X_2 + 2X_3 &\leq 40 \\ X_1 + 5X_2 + X_3 &\leq 45 \\ X_1, X_2, X_3 &\geq 0 \end{aligned}$$

TORA's optimum solution:

$$X_1 = 0, X_2 = 36, X_3 = 2, Z = \$882$$

Status of resources:

Resource	slack	status
Leather	0	scarce
Sewing	0	scarce
Finishing	25	abundant

Problem Set 3.4.A

Iteration	Basic	x_1	x_2	x_3	R_1	R_2	x_4	Solution
0 (starting)	z	$-4+7M$	$-1+4M$	$-M$	0	0	0	$9M$
x_1 enters	R_1	3	1	0	1	0	0	3
R_1 leaves	R_2	4	3	-1	0	1	0	6
	x_4	1	2	0	0	0	1	4
1	z	0	$\frac{1+5M}{3}$	$-M$	$\frac{4-7M}{3}$	0	0	$4+2M$
x_2 enters	x_1	1	$1/3$	0	$1/3$	0	0	1
R_1 leaves	R_2	0	$5/3$	-1	$-4/3$	1	0	2
	x_4	0	$5/3$	0	$-1/3$	0	1	3
2	z	0	0	$1/5$	$8/5-M$	$-1/5-M$	0	$18/5$
x_3 enters	x_1	1	0	$1/5$	$-3/5$	$-1/5$	0	$3/5$
x_4 leaves	x_2	0	1	$-3/5$	$-4/5$	$3/5$	0	$6/5$
	x_3	0	0	1	1	-1	1	1
3	z	0	0	0	$7/5-M$	$-M$	$-1/5$	$17/5$
(optimum)	x_1	1	0	0	$2/5$	0	$-1/5$	$2/5$
	x_2	0	1	0	$-1/5$	0	$3/5$	$9/5$
	x_3	0	0	1	1	-1	1	1

 $M=1:$

Optimum solution: $x_1=0, x_2=2, x_4=1$
 $Z=3$

Solution is infeasible because x_4 is positive. The reason $M=1$ produces an infeasible solution is that it does not play the role of a penalty relative to the objective coefficients of the real variables, x_1 and x_2 . Using $M=1$ makes x_4 more attractive than x_1 from the standpoint of minimizing.

 $M=10:$

Optimum solution: $x_1=0.4, x_2=1.8, Z=3.4$

The solution is feasible because it does not include artificials at positive level. $M=10$ is relatively much larger than the objective coefficients of x_1 and x_2 , and hence properly plays the role of a penalty.

 $M=1000:$

It produces the optimum solution as with $M=10$. The conclusion is that it suffices to select M reasonably larger than the objective coefficients of the real variables. Actually, $M=1000$ is an "overkill" in this case, and selecting such huge values could result in adverse roundoff error.

(a) Minimize $Z = 4x_1 + x_2 + M(R_1 + R_2 + R_3)$
 Subject to

$$\begin{aligned} 3x_1 + x_2 + R_1 &= 3 \\ 4x_1 + 3x_2 - S_2 + R_2 &= 6 \\ x_1 + 2x_2 - S_3 + R_3 &= 4 \\ x_1, x_2, S_2, S_3, R_1, R_2, R_3 &\geq 0 \end{aligned}$$

Basic	x_1	x_2	S_2	S_3	R_1	R_2	R_3	
Z	-4	-1			$(-M)$	$(-M)$	$(-M)$	0
R_1	3	1			1			3
R_2	4	3	-1			1		6
R_3	1	2		-1			1	4
Z	$-4+8M$	$-1+6M$	$-M$	$-M$	0	0	0	$10M$
R_1	3	1			1			3
R_2	4	3	-1			1		6
R_3	1	2		-1			1	4

(b) Minimize $Z = 4x_1 + x_2 + M R_1$
 Subject to

$$\begin{aligned} 3x_1 + x_2 + R_1 &= 3 \\ 4x_1 + 3x_2 + S_2 &= 6 \\ x_1 + 2x_2 + S_3 &= 4 \end{aligned}$$

Basic	x_1	x_2	R_1	S_2	S_3	
Z	-4	-1	$(-M)$			0
R_1	3	1	1			3
S_2	4	3		1		6
R_3	1	2			1	4
Z	$-4+3M$	$-1+M$	0	0	0	$3M$
R_1	3	1	1			3
S_2	4	3		1		6
R_3	1	2			1	4

(c) Minimize $Z = 4x_1 + x_2 + M(R_1 + R_2)$

$$\begin{aligned} 3x_1 + x_2 + R_1 &= 3 \\ 4x_1 + 3x_2 + R_2 &= 6 \\ x_1 + 2x_2 + S_3 &= 4 \end{aligned}$$

Basic	x_1	x_2	R_1	R_2	S_3	
Z	-4	-1	$(-M)$	$(-M)$	0	0
R_1	3	1	1			3
R_2	4	3		1		6
S_3	1	2			1	4
Z	$-4+7M$	$-1+4M$	0	0	0	$9M$
R_1	3	1	1			3
R_2	4	3		1		6
S_3	1	2			1	4

continued...

(d) Maximize $Z = 4x_1 + x_2 - M(R_1 + R_2)$

subject to

$$3x_1 + x_2 + R_1 = 3$$

$$4x_1 + 3x_2 - S_2 + R_2 = 6$$

$$x_1 + 2x_2 + S_3 = 4$$

Basic	x_1	x_2	S_2	R_1	R_2	S_3	
Z	-4	-1	0	(M)	(M)	0	0
R_1	3	1		(1)			3
R_2	4	3	-1		(1)		6
S_3	1	2				1	4
Z	-4-7M	-1-4M	M	0	0	0	-9M
R_1	3	1		1			3
R_2	4	3	-1		1		6
S_3	1	2				1	4

(a) Maximize $Z = 5x_1 + 6x_2 - M(R_1)$

subject to

$$-2x_1 + 3x_2 + (R_1) = 3 \quad (1)$$

$$x_1 + 2x_2 + S_3 = 5 \quad (3)$$

$$6x_1 + 7x_2 + S_4 = 3 \quad (4)$$

$$Z - (5-2M)x_1 - (6+3M)x_2 = -3M$$

(b) Maximize $Z = 2x_1 - 7x_2 - M(R_1 + R_2 + R_5)$

subject to

$$-2x_1 + 3x_2 + R_1 = 3 \quad (1)$$

$$4x_1 + 5x_2 - S_2 + R_2 = 10 \quad (2)$$

$$6x_1 + 7x_2 + S_4 = 3 \quad (4)$$

$$4x_1 + 8x_2 - S_5 + R_5 = 5 \quad (5)$$

$$Z - (2+6M)x_1 - (-7+16M)x_2 + MS_2 + MS_5 = -18M$$

(c) Minimize $Z = 3x_1 + 6x_2 + MR_5$

subject to

$$x_1 + 2x_2 + S_1 = 5 \quad (3)$$

$$6x_1 + 7x_2 + S_2 = 3 \quad (4)$$

$$4x_1 + 8x_2 - S_5 + R_5 = 5 \quad (5)$$

$$Z - (3-4M)x_1 - (6-8M)x_2 - MS_5 = 5M$$

(d) Minimize $Z = 4x_1 + 6x_2 + M(R_1 + R_2 + R_5)$

subject to

$$-2x_1 + 3x_2 + R_1 = 3 \quad (1)$$

$$4x_1 + 5x_2 - S_2 + R_2 = 10 \quad (2)$$

$$4x_1 + 8x_2 - S_5 + R_5 = 5 \quad (5)$$

$$Z - (4-6M)x_1 - (6-16M)x_2 - MS_2 - MS_5 = 18M$$

(e) Minimize $Z = 3x_1 + 2x_2 + M(R_1 + R_5)$

subject to

$$-2x_1 + 3x_2 + R_1 = 3 \quad (1)$$

$$4x_1 + 8x_2 - S_5 + R_5 = 5 \quad (5)$$

$$Z - (3-2M)x_1 - (2-11M)x_2 - MS_5 = 8M$$

continued...

(a)

5

Basic	x_1	x_2	x_3	S_2	R_1	R_2	
Z	-2	-3	5	M	0	0	-17M
R_1	1	1	1	0	1	0	7
R_2	(2)	-5	1	-1	0	1	10
Z	0	-8	6	-1	0	1	10-2M
R_1	0	$-\frac{7M}{2}$	$-\frac{M}{2}$	$-\frac{M}{2}$	0	$+\frac{3M}{2}$	
x_1	1	$-\frac{5}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	5
Z	0	0	$50/7$	$1/7$	$16/7$	$-1/7$	$\frac{102}{7}$
x_2	0	1	$1/7$	$1/7$	$2/7$	$-1/7$	$4/7$
x_1	1	0	$6/7$	$-1/7$	$5/7$	$1/7$	$45/7$

(b)

Basic	x_1	x_2	x_3	S_2	R_1	R_2	Soln
Z	-2	-3	5	-M	0	0	17M
R_1	1	1	1	0	1	0	7
R_2	2	-5	1	-1	0	1	10
Z	0	-8	6	-1	0	1	$10+2M$
R_1	0	$7/2$	$1/2$	$1/2$	1	$-1/2$	2
x_1	1	$-5/2$	$1/2$	$-1/2$	0	$1/2$	5
Z	0	0	$50/7$	$1/7$	$16/7$	$-1/7$	$\frac{102}{7}$
x_2	0	1	$1/7$	$1/7$	$2/7$	$-1/7$	$4/7$
x_1	1	0	$6/7$	$-1/7$	$5/7$	$1/7$	$45/7$
Z	0	-50	0	-7	-12	7	-14
x_3	0	7	1	1	2	-1	4
x_1	1	-6	0	-1	-1	1	3

continued...

(c)

Basic	x_1	x_2	x_3	S_2	R_1	R_2	Sol ⁿ
Z	-1	-2	-1	0	M	M	-
R_1	1	1	1	0	1	0	7
R_2	2	-5	1	-1	0	1	10
Z	-1	-2	-1	M	0	0	-17M
R_1	1	1	1	0	1	0	7
R_2	2	-5	1	-1	0	1	10
Z	0	$-\frac{9}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	$-\frac{17M}{2}$
R_1	0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	$-\frac{1}{2}$	2
x_1	1	$-\frac{5}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	5
Z	0	0	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$-\frac{1}{7}$	$\frac{53}{7}$
x_2	0	1	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{2}{7}$	$-\frac{1}{7}$	$\frac{41}{7}$
x_1	1	0	$\frac{6}{7}$	$-\frac{1}{7}$	$\frac{5}{7}$	$\frac{1}{7}$	$\frac{45}{7}$

(d)

Basic	x_1	x_2	x_3	S_2	R_1	R_2	Sol ⁿ
Z	-4	8	-3	0	-M	-M	0
R_1	1	1	1	0	1	0	7
R_2	2	-5	1	-1	0	1	10
Z	-4	8	-3	-M	0	0	17M
R_1	1	1	1	0	1	0	7
R_2	2	-5	1	-1	0	1	10
Z	0	$-\frac{2}{7}$	$-\frac{1}{7}$	$-\frac{2}{7}$	0	$\frac{2}{7}$	$-\frac{20}{7}$
R_1	0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	$-\frac{1}{2}$	2
x_1	1	$-\frac{5}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	5
Z	0	0	$-\frac{5}{7}$	$-\frac{12}{7}$	$\frac{4}{7}$	$\frac{14}{7}$	$\frac{148}{7}$
x_2	0	1	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{2}{7}$	$-\frac{1}{7}$	$\frac{41}{7}$
x_1	1	0	$\frac{6}{7}$	$-\frac{1}{7}$	$\frac{5}{7}$	$\frac{1}{7}$	$\frac{45}{7}$

In the first iteration, we must substitute out the starting solution variables, x_3 and x_4 , in the Z-equation, exactly as we do with the artificial variables

6

Basic	x_1	x_2	x_3	x_4	Solution
Z	-2	-4	(-4)	(3)	-
x_3	1	1	(1)	0	4
x_4	1	4	0	(1)	8
Z	-1	-12	0	0	-8
x_3	1	1	1	0	4
x_4	1	(4)	0	1	8
Z	2	0	0	3	16
x_3	$\frac{3}{4}$	0	1	$-\frac{1}{4}$	2
x_2	$\frac{1}{4}$	1	0	$\frac{1}{4}$	2

After adding surplus S_1 and S_2 , substitute out x_3 in the Z-equation

7

Basic	x_1	x_2	S_1	S_2	x_3	x_4	Solution
Z	-3	-2	0	0	(-3)	0	-
x_3	1	4	-1	0	(1)	0	7
x_4	2	1	0	-1	0	1	10
Z	0	10	-3	0	0	0	21
x_3	1	4	-1	0	1	0	7
x_4	2	1	0	-1	0	1	10
Z	$-\frac{5}{2}$	0	$-\frac{1}{2}$	0	$-\frac{5}{2}$	0	$\frac{7}{2}$
x_2	$\frac{1}{4}$	1	$-\frac{1}{4}$	0	$\frac{1}{4}$	0	$\frac{7}{4}$
x_4	$\frac{7}{4}$	0	$\frac{1}{4}$	-1	$-\frac{1}{4}$	1	$\frac{33}{4}$

Both x_3 and R (the starting solution variables) must be substituted out in the Z-equation

8

Basic	x_1	x_2	x_3	R	Solution
Z	-1	-5	(-3)	(M)	-
x_3	1	2	(1)	0	3
R	2	-1	0	(1)	4
Z	$2-2M$	$1+M$	0	0	$9-4M$
x_3	1	2	1	0	3
R	(2)	-1	0	1	4
Z	0	2	0	$-1+M$	5
x_3	0	$\frac{5}{2}$	1	$-\frac{1}{2}$	1
x_1	1	$-\frac{1}{2}$	0	$\frac{1}{2}$	2

9

Maximize $Z = 2x_1 + 5x_2 - MR_1$

Subject to

$$3x_1 + 2x_2 - S_1 + R_1 = 6$$

$$2x_1 + x_2 + S_2 = 2$$

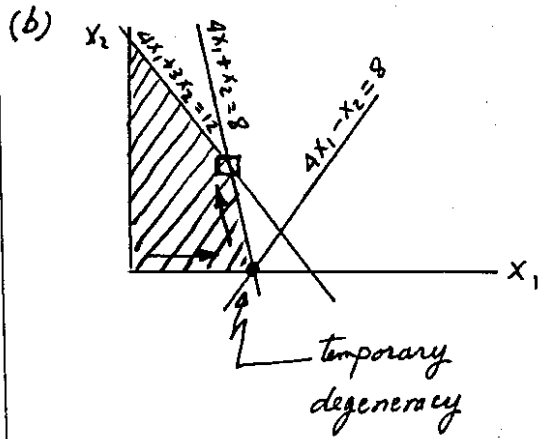
$$x_1, x_2, S_1, R_1, S_2 \geq 0$$

Basic	x_1	x_2	S_1	R_1	S_2	
Z	-2	-5	0	M	0	-
R_1	3	2	-1	1	0	6
S_2	2	1	0	0	1	2
Z	$-2-3M$	$-5-2M$	M	0	0	$-6M$
R_1	3	2	-1	1	0	6
S_2	2	1	0	0	1	2
Z	0	$-4-M/2$	M	0	$1+3M/2$	$-2+3M$
R_1	0	$1/2$	-1	1	$-3/2$	3
x_1	1	$1/2$	0	0	$1/2$	1
Z	$8+M$	0	M	0	$5+2M$	$10-2M$
R_1	-1	0	-1	1	-2	2
x_2	2	1	0	0	1	2

The Z-row shows that the solution is optimal (all nonbasic coefficients in the Z-row are ≥ 0). However, the solution is infeasible because the artificial variable R_1 assumes a positive value. Having a positive value for the artificial variable R_1 is the same as regarding the constraint $3x_1 + 2x_2 \geq 6$ as $3x_1 + 2x_2 \leq 6$, which violates the constraints of the original model.

Problem Set 3.5.A

(a) From TORA, iterations 2 and 3 are degenerate. Degeneracy is removed in iteration 4. 2



Problem Set 3.5.B

1							
Basic	x_1	x_2	x_3	s_1	s_2	s_3	Solution
Z	-1	-2	-3	0	0	0	0
s_1	1	2	3	1	0	0	10
s_2	1	1	0	0	1	0	5
s_3	1	0	0	0	0	1	1
Z	0	0	0	1	0	0	10
x_3	1/3	2/3	1	1/3	0	0	10/3
s_2	1	1	0	0	1	0	5
s_3	1	0	0	0	0	1	1
Z	0	0	0	1	0	0	10
x_3	-1/3	0	1	1/3	-2/3	0	0
x_2	1	1	0	0	1	0	5
s_3	1	0	0	0	0	1	1
Z	0	0	0	1	0	0	10
x_3	0	0	1	1/3	-2/3	1/3	1/3
x_2	0	1	0	0	1	-1	4
x_1	1	0	0	0	0	1	1
Z	0	0	0	1	0	0	10
x_3	0	2/3	1	1/3	0	-1/3	3
s_3	0	1	0	0	1	-1	4
x_1	1	0	0	0	0	1	1

Three alternative basic optima:

$$(x_1, x_2, x_3) = \begin{cases} (0, 0, 10/3) \\ (0, 5, 0) \\ (1, 4, 1/3) \end{cases}$$

The associated nonbasic alternative optima are

$$\begin{aligned} \tilde{x}_1 &= \lambda_3 \\ \tilde{x}_2 &= 5\lambda_2 + 4\lambda_3 \\ \tilde{x}_3 &= 10/3\lambda_1 + 1/3\lambda_3 \end{aligned}$$

where

$$\lambda_1 + \lambda_2 + \lambda_3 = 1$$

$$0 \leq \lambda_i \leq 1, i = 1, 2, 3$$

2						
Basic	x_1	x_2	x_3	s_1	s_2	
Z	-2	1	3	0	0	0
s_1	1	-1	5	1	0	10
s_2	2	-1	3	0	1	40
Z	-7/5	2/5	0	3/5	0	6
x_3	1/5	-1/5	1	1/5	0	2
s_2	7/5	-2/5	0	-3/5	1	34
Z	0	-1	7	2	0	20
x_1	1	-1	5	1	0	10
s_2	0	1	-7	-2	1	20
Z	0	0	0	0	1	40
x_1	1	0	-2	-1	0	30
x_2	0	1	-7	-2	1	20

x_3 and s_1 can yield alternative optima. However, because all their constraint coefficients are negative (in general, ≤ 0), none can yield an alternative (corner point) basic solution.



3							
Basic	x_1	x_2	x_3	s_1	s_2	s_3	Solution
Z	-3	-1	0	0	0	0	0
s_1	1	2	0	1	0	0	5
s_2	1	1	-1	0	1	0	2
s_3	7	3	-5	0	0	1	20
Z	0	2	-3	0	3	0	6
s_1	0	1	1	1	-1	0	3
x_1	1	1	-1	0	1	0	2
s_3	0	-4	2	0	-7	1	6
Z	0	5	0	3	0	0	15
x_3	0	1	1	1	-1	0	3
x_1	1	2	0	1	0	0	5
s_3	0	-6	0	-2	-5	1	0

The optimum solution is degenerate because s_3 is basic and equal to zero. Also, it has alternative nonbasic solutions because s_2 has a zero coefficient in the Z -row and all its constraint coefficients are ≤ 0 .

Problem Set 3.5.C

Basic	x_1	x_2	s_1	s_2	
Z	-2	-1	0	0	0
s_1	1	-1	1	0	10
s_2	2	0	0	1	40
Z	0	-3	2	0	20
x_1	1	-1	1	0	10
s_2	0	2	-2	1	20
Z	0	0	-1	3/2	50
x_1	1	0	0	1/2	20
x_2	0	1	-1	1/2	10

unbounded $\rightarrow$

(a)

x_2
-10
-3
0
-1

$\Rightarrow$ solution space unbounded
in the direction of x_2

(b) Objective value is unbounded
because each unit increase
in x_2 increases Z by 10

Problem Set 3.5.D

x_1 = number of units of T1
 x_2 = number of units of T2
 x_3 = number of units of T3

Constraints:

$$3x_1 + 5x_2 + 6x_3 \leq 1000$$

$$5x_1 + 3x_2 + 4x_3 \leq 1200$$

$$x_1 + x_2 + x_3 \geq 500$$

$$x_1, x_2, x_3 \geq 0$$

We can use Phase I to see whether the problem has a feasible solution; that is,

minimize $r = R_3$

subject to

$$3x_1 + 5x_2 + 6x_3 + S_1 = 1000$$

$$5x_1 + 3x_2 + 4x_3 + S_2 = 1200$$

$$x_1 + x_2 + x_3 - S_3 + R_3 = 500$$

$$x_1, x_2, x_3, S_1, S_2, S_3, R_3 \geq 0$$

Optimum solution from TORA:

$$R_3 = r = 225 \text{ units}$$

This is interpreted as a deficiency of 225 units. The most that can be produced is $500 - 225 = 275$ units

1								2							
Basic	x_1	x_2	x_3	S_1	S_2	R_1	Sol'n								
Z	-3	-2	-3												
	-3M	-4M	-2M	M	0	0	-8M								
S_1	2	1	1	0	1	0	2								
R_1	3	4	2	-1	0	1	8								
Z	-1	-1	-1		2										
	+5M	0	+2M	M	+4M	0	4								
x_2	2	1	1	0	1	0	2								
R_1	-5	0	-2	-1	-4	1	0								

Because $R_1 = 0$ in the optimal tableau, the problem has a feasible solution. The optimum solution is

$$x_1 = 0, x_2 = 2, Z = 4$$

Note that in the first iteration, R_1 could have been used as the leaving variable, in which case it would not be basic in the optimum iteration.

Problem Set 3.6.A

 $X_1 = \text{Nbr. units of product A}$
 $X_2 = \text{Nbr. units of product B}$

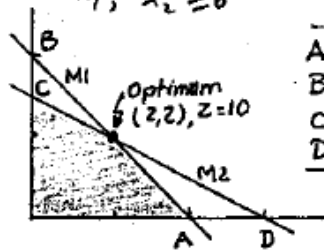
 Maximize $Z = 2X_1 + 3X_2$

s.t.

$$2X_1 + 2X_2 \leq 8 \quad (M1)$$

$$3X_1 + 6X_2 \leq 18 \quad (M2)$$

$$X_1, X_2 \geq 0$$



	M1	M2	Z
A = (4,0)		12	8
B = (0,4)		24	12
C = (0,3)	6		9
D = (6,0)	12		12

(a) $M1 \text{ at } C = 2(0) + 2(3) = 6$

$$M1 \text{ at } D = 2(6) + 2(0) = 12$$

$$Z \text{ at } C = 2(0) + 3(3) = 9$$

$$Z \text{ at } D = 2(6) + 3(0) = 12$$

$$\text{Dual price} = \frac{12-9}{12-6} = \$0.50/\text{unit}$$

$$\text{Allowable range} = (6 \leq M1 \leq 12)$$

$$M2 \text{ at } A = 3(4) + 6(0) = 12$$

$$M2 \text{ at } B = 3(0) + 6(4) = 24$$

$$Z \text{ at } A = 2(4) + 3(0) = 8$$

$$Z \text{ at } B = 2(0) + 3(4) = 12$$

$$\text{Dual price} = \frac{12-8}{24-12} = \$0.33/\text{unit}$$

$$\text{Range: } 12 \leq M2 \leq 24$$

 (b) Dual price = \$0.50/unit valid in the range $6 \leq M1 \leq 12$

$$\text{Increase in revenue} = 0.5 \times 4 = \$2.00$$

$$\text{Increase in cost} = 0.3 \times 4 = \$1.20$$

Cost < Revenue -- purchase recommended

 (c) Dual price = \$0.33/unit valid in the range $12 \leq M2 \leq 24$

$$\text{Purchase price/unit} < 0.33$$

 (d) Dual price = \$0.33/unit valid in the range $12 \leq M2 \leq 24$. $M2$ is increased from 18 to 23 units

Increase in revenue

$$= 5 \times 0.33 = \$1.65$$

$$\text{New optimum revenue} = 10 + 1.65 = \$11.65$$

Problem Set 3.6B

1

(a) $\frac{3}{6} \leq \frac{C_A}{C_B} \leq \frac{2}{2}$, or
 $.5 \leq \frac{C_A}{C_B} \leq 1$ or $1 \leq \frac{C_B}{C_A} \leq 2$

(b) Maximize $Z = 2x_A + 3x_B$
 $C_B = 3$: $3 \times .5 \leq C_A \leq 3 \times 1$
 $1.5 \leq C_A \leq 3$
 $C_A = 2$: $2 \times .5 \leq C_B \leq 2 \times 2$
 $1 \leq C_B \leq 4$

(c) $\frac{C_A}{C_B} = \frac{5}{4} = 1.25$, which falls outside the range $.5 \leq \frac{C_A}{C_B} \leq 1$. Optimum solution changes and must be computed anew.
 New solution: $x_A = 4$, $x_B = 0$, $Z = \$20$.

(d) Case 1: $Z = 5x_A + 3x_B$
 $C_A = 5$ falls outside the range $(1.5, 3)$, hence the optimum changes. New Optimum is $x_A = 4$, $x_B = 0$, $Z = \$20$.
Case 2: $Z = 2x_A + 4x_B$
 $C_B = 4$ falls in the range $(1, 4)$, hence optimum is unchanged at $x_A = x_B = 2$,
 $Z = 2(2) + 4(2) = 12$

Problem Set 3.6C

Feasibility conditions:

$$x_2 = 100 + \frac{1}{2} D_1 - \frac{1}{4} D_2$$

$$x_3 = 230 + \frac{1}{2} D_2$$

$$x_6 = 20 - 2 D_1 + D_2 + D_3$$

$$(a) D_1 = 438 - 430 = 8 \text{ min}$$

$$D_2 = 500 - 460 = 40$$

$$D_3 = 410 - 420 = -10$$

$$x_2 = 100 + \frac{1}{2}(8) - \frac{1}{4}(40) = 94 > 0$$

$$x_3 = 230 + \frac{1}{2}(40) = 250 > 0$$

$$x_6 = 20 - 2(8) + 40 - 10 = 34 > 0$$

Dual prices:

$$\text{Resource 1} = \$1/\text{min}, -200 \leq D_1 \leq 10$$

$$2 = \$2/\text{min}, -20 \leq D_2 \leq 400$$

$$3 = \$0/\text{min}, -20 \leq D_3 < \infty$$

$$\begin{aligned} \text{New profit} &= 1350 + D_1 + 2D_2 + 0D_3 \\ &= 1350 + 8 + 2 \times 40 = 1438 \end{aligned}$$

$$(b) D_1 = 460 - 430 = 30 \text{ min}$$

$$D_2 = 440 - 460 = -20$$

$$D_3 = 380 - 420 = -40$$

$$x_2 = 100 + \frac{1}{2}(30) - \frac{1}{4}(-20) = 120 > 0$$

$$x_3 = 230 + \frac{1}{2}(-20) = 220 > 0$$

$$x_6 = 20 - 2(30) - 20 - 40 = -100 < 0$$

3

 X_i = units of product $i = 1, 2, 3$ Maximize $Z = 20X_1 + 50X_2 + 35X_3$

s.t.

$$-0.5X_1 + 0.5X_2 + 0.5X_3 \leq 0$$

$$X_1 \leq 75$$

$$2X_1 + 4X_2 + 3X_3 \leq 240$$

$$X_1, X_2, X_3 \geq 0$$

(a) Solution: $Z = \$2800$

$$X_1 = X_2 = 40, X_3 = 0$$

	X_1	X_2	X_3	S_1	S_2	S_3	
Z	0	0	$10/3$	$20/3$	0	$35/3$	2800
X_2	0	0	$5/6$	$2/3$	0	$1/6$	40
S_2	1	0	$1/6$	$4/3$	1	$-1/6$	35
X_1	0	1	$-1/6$	$-4/3$	0	$1/6$	40

$$(b) Z + 10/3 X_3 + 20/3 S_1 + 0 S_2 + 35/3 S_3 = 2800$$

Dual-price for raw material = $\$35/3$ / lb

$$\left. \begin{array}{l} X_2 = 40 + D_3/6 \\ S_2 = 35 - D_3/6 \\ X_1 = 40 + D_3/6 \end{array} \right\} \Rightarrow -240 \leq D_3 \leq 210$$

 $D_3 = 120/6$ falls in the range $(-240, 210)$

New solution:

$$X_1 = 40 + \frac{120}{6} = 60 \text{ units}$$

$$X_2 = 40 + \frac{120}{6} = 60 \text{ units}$$

$$X_3 = 0$$

$$\begin{aligned} \text{New revenue} &= 2800 + (35/3)(120) \\ &= \$4200 \end{aligned}$$

continued...

(c) Dual price = 0, $-35 \leq D_1 < \infty$

$$\pm 10\% \text{ of } 75 = \pm 7.5 \text{ or}$$

Change has no effect on the solution

Problem Set 3.6D

From Section 3.6.3, we have the following optimality conditions for the TOYCO model:

$$x_1: 4 - \frac{1}{4}d_2 + \frac{3}{2}d_3 - d_1 \geq 0$$

$$x_4: 1 + \frac{1}{2}d_2 \geq 0$$

$$x_5: 2 - \frac{1}{4}d_2 + \frac{1}{2}d_3 \geq 0$$

$$(i) Z = 2x_1 + x_2 + 4x_3$$

$$d_1 = 2 - 3 = -1, d_2 = 1 - 2 = -1, d_3 = 4 - 5 = -1$$

$$x_1: 4 - \frac{1}{4}(-1) + \frac{3}{2}(-1) - (-1) = 3.75 > 0$$

$$x_4: 1 + \frac{1}{2}(-1) = .5 > 0$$

$$x_5: 2 - \frac{1}{4}(-1) + \frac{1}{2}(-1) = 1.75 > 0$$

Conclusion: Solution is unchanged

$$(ii) Z = 3x_1 + 6x_2 + x_3$$

$$d_1 = 3 - 3 = 0, d_2 = 6 - 2 = 4, d_3 = 1 - 5 = -4$$

$$x_1: 4 - \frac{1}{4}(4) + \frac{3}{2}(-4) - (0) = -3 < 0$$

Conclusion: solution changes

$$(iii) Z = 8x_1 + 3x_2 + 9x_3$$

$$d_1 = 8 - 3 = 5, d_2 = 3 - 2 = 1, d_3 = 9 - 5 = 4$$

$$x_1: 4 - \frac{1}{4}(1) + \frac{3}{2}(4) - (5) = 4.75 > 0$$

$$x_4: 1 + \frac{1}{2}(1) = 1.5 > 0$$

$$x_5: 2 - \frac{1}{4}(1) + \frac{1}{2}(4) = 3.75 > 0$$

Conclusion: solution is unchanged

2

x_1 = Nbr cans of A1
 x_2 = Nbr. cans of A2
 x_3 = Nbr. cans of BK

Maximize $Z = 80x_1 + 70x_2 + 60x_3$

s.t. $x_1 + x_2 + x_3 \leq 500 \quad \leftarrow S_1$
 $x_1 \geq 100 \quad \leftarrow S_2$
 $4x_1 - 2x_2 - 2x_3 \leq 0 \quad \leftarrow S_3$
 $x_1, x_2, x_3 \geq 0$

TORA optimum tableau:

Basic	x_1	x_2	x_3	S_1	S_2	S_3	Solution
Z	0	0	10	73.33	0	1.67	3666.67
x_2	0	1	1	.67	0	-.17	333.33
x_1	1	0	0	.33	0	.17	166.67
S_2	0	0	0	.33	1	-.17	66.67

continued...

(a) $Z = \$366.67$

$x_1 = 166.67, x_2 = 333.33, x_3 = 0$

(b) Reduced cost for $x_3 = 10$ cents. Price should be increased by more than 10 cents/can.

(c) $d_1 = d_2 = d_3 = -5$ cents

From the optimum tableau, reduced costs:

$$x_3: 10 + d_2 - d_3 = 10 - 5 - (-5) = 10 > 0$$

$$S_1: 73.33 + .67d_2 + .33d_1 = 73.33 + .67(-5) + .33(-5) = 68.33 > 0$$

$$S_3: 1.67 - .17d_2 + .17d_1 = 1.67 - .17(-5) + .17(-5) = 1.67 > 0$$

conclusion: solution is unchanged.

Problem Set 4.1.A

Primal:

Minimize $Z = 5x_1 + 12x_2 + 4x_3$
 Subject to

$$\begin{aligned} x_1 + 2x_2 + x_3 + s_1 &= 10 \\ 2x_1 - x_2 + 3x_3 &= 8 \\ x_1, x_2, x_3, s_1 &\geq 0 \end{aligned}$$

Dual:

Maximize $w = 10y_1 + 8y_2$
 Subject to

$$\begin{aligned} y_1 + 2y_2 &\leq 5 \\ 2y_1 - y_2 &\leq 12 \\ y_1 + 3y_2 &\leq 4 \\ y_1 &\leq 0 \\ y_2 &\text{ unrestricted} \end{aligned}$$

Primal:

Minimize $Z = 15x_1 + 12x_2$
 Subject to

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 3 \\ 2x_1 - 4x_2 + x_4 &= 5 \\ 3x_1 + x_2 &= 4 \\ x_1, x_2, x_3, x_4 &\geq 0 \end{aligned}$$

Dual:

Maximize $Z = 3y_1 + 5y_2 + 4y_3$
 Subject to

$$\begin{aligned} y_1 + 2y_2 + 3y_3 &\leq 15 \\ 2y_1 - 4y_2 + y_3 &\leq 12 \\ -y_1 &\leq 0 \Rightarrow y_1 \geq 0 \\ y_2 &\leq 0 \\ y_3 &\text{ unrestricted} \end{aligned}$$
Primal:

Minimize $Z = 5x_1^+ - 5x_1^- + 6x_2$
 Subject to

$$\begin{aligned} x_1^+ - x_1^- + 2x_2 &= 5 \\ -x_1^+ + x_1^- + 5x_2 - x_3 &= 3 \\ 4x_1^+ - 4x_1^- + 7x_2 + x_4 &= 8 \\ x_1^+, x_1^-, x_2, x_3, x_4 &\geq 0 \end{aligned}$$

Dual:

Maximize $Z = 5y_1 + 3y_2 + 8y_3$
 Subject to

$$\begin{aligned} y_1 - y_2 + 4y_3 &\leq 5 \\ -y_1 + y_2 - 4y_3 &\leq -5 \Rightarrow y_1 - y_2 + 4y_3 = 5 \\ 2y_1 + 5y_2 + 7y_3 &\leq 6 \\ -y_2 &\leq 0 \Rightarrow y_2 \geq 0 \\ y_3 &\leq 0 \\ y_1 &\text{ unrestricted} \end{aligned}$$
(a) Primal:

Maximize $Z = -5x_1 + 2x_2$
 s.t.

$$\begin{aligned} x_1 - x_2 - x_3 &= 2 \\ 2x_1 + 3x_2 + x_4 &= 5 \\ x_1, x_2, x_3, x_4 &\geq 0 \end{aligned}$$

Dual:

Minimize $w = 2y_1 + 5y_2$
 Subject to

$$\begin{aligned} y_1 + 2y_2 &\geq -5 \\ -y_1 + 3y_2 &\geq 2 \\ -y_1 &\geq 0 \Rightarrow y_1 \leq 0 \\ y_2 &\geq 0 \end{aligned}$$
(b) Primal:

Minimize $Z = 6x_1 + 3x_2$
 Subject to

$$\begin{aligned} 6x_1 - 3x_2 + x_3 - x_4 &= 2 \\ 3x_1 + 4x_2 + x_3 - x_5 &= 5 \\ x_1, x_2, x_3, x_4, x_5 &\geq 0 \end{aligned}$$

Dual:

Maximize $w = 2y_1 + 5y_2$
 Subject to

$$\begin{aligned} 6y_1 + 3y_2 &\leq 6 \\ -3y_1 + 4y_2 &\leq 3 \\ y_1 + y_2 &\leq 0 \\ -y_1 &\leq 0 \Rightarrow y_1 \geq 0 \\ -y_2 &\leq 0 \Rightarrow y_2 \geq 0 \end{aligned}$$
(c) Primal:

Maximize $Z = x_1 + x_2$
 Subject to

$$\begin{aligned} 2x_1 + x_2 &= 5 \\ 3x_1 - x_2 &= 6 \\ x_1, x_2 &\text{ unrestricted} \end{aligned}$$

Dual:

Minimize $w = 5y_1 + 6y_2$
 Subject to

$$\begin{aligned} 2y_1 + 3y_2 &= 1 \\ y_1 - y_2 &= 1 \\ y_1, y_2 &\text{ unrestricted} \end{aligned}$$

Problem Set 4.2.C

Dual: Maximize $w = 50y$

s.t.

$$5y_1 \leq 10, -7y_1 \leq 4, 3y_1 \leq 5, y_1 \geq 0$$

The constraints simplify to

$$0 \leq y_1 \leq 5/3$$

$$\text{Thus, } \max w = 50 \times \frac{5}{3} = \frac{250}{3} = \min z$$

Dual:

$$\text{Maximize } w = 50y_1 + 20y_2 + 30y_3 + 35y_4 + 10y_5 + 90y_6 + 20y_7$$

s.t.

$$5y_1 + y_2 + 7y_3 + 5y_4 + 2y_5 + 12y_6 \leq 5$$

$$5y_1 + y_2 + 6y_3 + 5y_4 + 4y_5 + 10y_6 + y_7 \leq 6$$

$$3y_1 - y_2 - 9y_3 + 5y_4 - 15y_5 - 10y_7 \leq 3$$

$$-y_j \leq 0 \Rightarrow y_j \geq 0, j = 1, 2, \dots, 7$$

From TORA, optimal objective equation is

$$z + 50y_1 + 0y_2 + 90y_3 + 65y_4 + 70y_5 + 40y_6 + 0y_7 + 0s_1 + 20s_2 + 0s_3 = 120$$

(s_1, s_2, s_3) are slack variables.

$$\text{Thus, } x_1 = 0, x_2 = 20, x_3 = 0$$

Obtaining the solution from the dual is advantageous computationally because the dual has a smaller number of constraints.

Problem Set 4.4.A

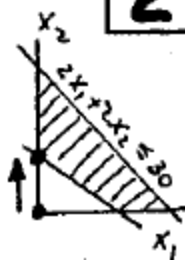
(a)

Minimize $Z = 2x_1 + 3x_2$
 subject to

$$2x_1 + 2x_2 \leq 30$$

$$-x_1 - 2x_2 \leq -10$$

$$x_1, x_2 \geq 0$$



2

Basic	x_1	x_2	x_3	x_4	S_{01}
Z	-2	-3	0	0	0
x_3	2	2	1	0	30
x_4	-1	-2	0	1	-10
Z	-1/2	0	0	-3/2	15
x_3	1	0	1	1	20
x_2	1/2	1	0	-1/2	5

(d)

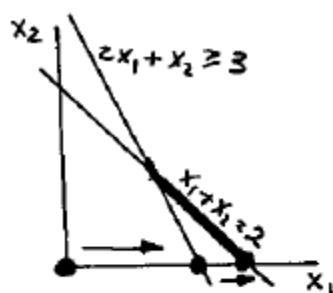
Minimize $Z = 2x_1 + 3x_2$
 subject to

$$2x_1 + x_2 \geq 3$$

$$x_1 + x_2 \leq 2$$

$$x_1 + x_2 \geq 2$$

$$x_1, x_2 \geq 0$$



Basic	x_1	x_2	x_3	x_4	x_5	S_{01}
Z	-2	-3	0	0	0	0
x_3	-2	-1	1	0	0	-3
x_4	1	1	0	1	0	2
x_5	-1	-1	0	0	1	-2
Z	0	-2	-1	0	0	3
x_1	1	1/2	-1/2	0	0	3/2
x_4	0	1/2	1/2	1	0	1/2
x_5	0	-1/2	-1/2	0	1	-1/2
Z	0	-1	0	0	-2	4
x_1	1	1	0	0	-1	2
x_4	0	0	0	1	1	0
x_3	0	1	1	0	-2	1

Problem Set 5.1.A

- (a) False
(b) True
(c) True

1

(a) $\sum a_i = 25$, $\sum b_j = 31$

2

Add a dummy source whose supply amount is $31 - 25 = 6$ units

(b) $\sum a_i = 74$, $\sum b_j = 65$

Add a dummy destination whose demand amount is $74 - 65 = 9$ units

3

Denver will be 150 cars short.
Similarly, Miami will be 50 cars short of satisfying its demand

4

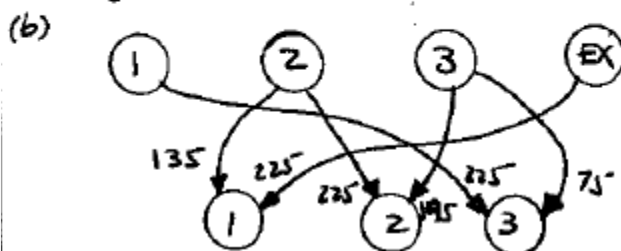
Assign a very high cost M to the route from Detroit to Dummy

Assume units in 100,000 kWh

7

(a)

	city 1	2	3	
Plant	60	70	40	
1			225	225
2	32	30	35	360
3	50	48	45	270
EX	100	100	M	225
	360	420	300	



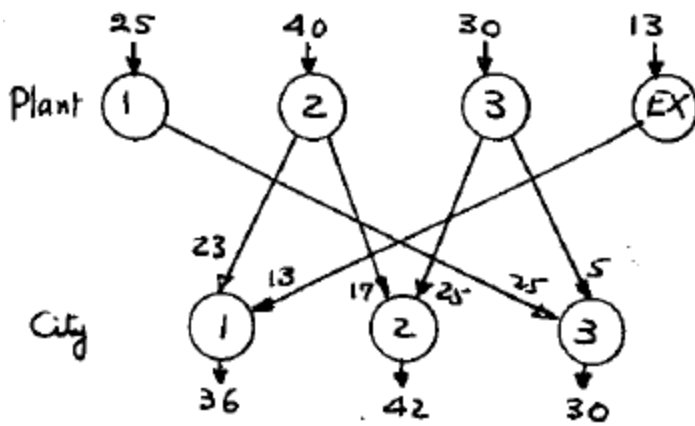
(c) City 1 excess cost = \$22,500

Optimum cost = \$55,305*

(a)

	city			
	1	2	3	
Plant 1	600 25	700 17	400 25	25
Plant 2	320 23	300 17	350 5	40
Plant 3	500 13	480 25	450 5	30
Excess plant 4	1000 13	1000 13	M 13	13
	36	42	30	

6

(b) $M = \$10,000$ in TORA

Total cost = \$49,710

(c) City 1 excess cost = 13×1000
 $= \$13,000$

Problem Set 5.3.A

(a)

Northwest:

5	1		6
	4	3	7
		7	7
5	5	10	

Cost = \$42

Least-cost:

5		1	6
	5	2	7
		7	7
5	5	10	

Cost = \$37

Vogel:

5		1	6	Penalty	Penalty
	5		5	1	4
		7	7	1	1
5	5	10			

Cost = \$37

Penalty (2) 1 2 ← Step 1

Penalty - 1 2 ← Step 2

(b)

Northwest:

7		2	6	7
3	9		2	12
	1	10	5	11
10	10	10		

Cost = \$94

Least-Cost:

		7	6	7
10		2	12	
	10	1	11	
10	10	10		

Cost = \$61

continued...

VAM:

7		2	6	Penalties
2		4	10	2 4 -
1	10		5	2 2 2
1	1			

Cost = \$40

Penalties { 1 1 3 } Step 1

Penalties { 2 1 - } Step 2

Penalties { 2 1 - } Step 3

(c)

Northwest:

9	3		8	12
	7	4	7	14
		6	4	4
9	10	11		

Cost = \$104

Least-Cost:

2	10		8	12
3		4	11	14
4		6	7	4
9	10	11		

Cost = \$38

VAM:

2	10		8	Penalties
3		4	11	12 4 4
4		6	7	2 2
9	10	11		3 3

Cost = \$38

Penalties { 1 3 7 } Step 1

Penalties { 1 3 - } Step 2

continued...

Problem Set 5.3.B

Set 5.3

(i)

u \ v	0	2	6	
0	5	1	5	6
-1	-3	4	5	9
-3	-5	-5	5	5
	5	5	10	

u \ v	0	-3	1	
0	5	-5	1	6
4	2	5	4	9
7	0	-5	5	5
	5	5	10	

u \ v	0	-1	1	
0	1	-3	5	6
2	4	5	-2	9
2	0	-3	5	5
	5	5	10	

Cost = \$33

Alternative solution exists

(ii)

u \ v	0	4	2	
0	7	1	0	8
-1	-3	5	-3	5
-2	-3	0	6	6
	7	6	6	

Problem has alternative optima. Cost = \$19
 Note: If x_{23} were selected as the zero in place of x_{32} , solution would require one more iteration.

(iii)

u \ v	M	M-3	M-5	
0	4	3	5	4
7-M	1	4	-7	9
11-M	1	8	19	6
	5	6	19	

u \ v	6	3	1	
0	6-M	4	-4	4
1	5	2	-7	9
5	10	0	19	6
	5	6	19	

u \ v	6	3	11	
0	6-M	4	6	4
1	5	2	3	9
-5	0	-10	19	6
	5	6	19	

u \ v	0	-3	5	
0	-M	-6	4	4
7	1	6	3	9
1	4	-10	15	6
	5	6	19	

u \ v	0	0	5	
0	-M	-3	4	4
-3	-3	6	1	9
5	1	-7	14	6
	5	6	19	

Cost = \$142

(c)

Method	Nbr. of iterations		
	(i)	(ii)	(iii)
NW	3	4	5
Least cost	2	2	2
Vogel	2	1	1

Least-cost starting solution:

2

u \ v	2	1	2	
0	5 -3	1 (10)	7 -5	10
4	6 (70)	4 1	6 (10)	80
1	3 (5)	2 (10)	5 -2	15
0	5 -3	3 -2	2 (40)	40
	75	20	50	

u \ v	3	1	3	
0	5 -2	1 (10)	7 -4	10
3	6 (60)	4 (10)	6 (10)	80
0	3 (15)	2 -1	5 -2	15
-1	5 -3	3 -3	2 (40)	40
	75	20	50	

Destination 3 will be 40 units short. Optimum cost = \$595